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Silent imperatives: a multimodal approach to warning expressions

Abstract

This article argues in favor of a formal model that views language as an integrated multimodal system where syntax, prosody, and gestures are linked. It demonstrates that specific syntactic structures trigger sensorimotor realizations, which include intonation, phonology, and gestures. We argue that gestures, in particular non-lexical co-speech gestures, are not merely an additional channel but a fundamental part of grammar, integrated into the same system that governs word order and phonological realization. We suggest a syntactic representation of warning expressions that combines syntactic and pragmatic aspects, based on Italian data and experimental evidence. Specifically, we argue that warnings always include a warning call, which can be either lexical in content or expressed solely through special prosody and gestures.

Keywords: Warning expressions, co-speech gestures, multimodality, imperatives, generative grammar, syntax.

1 Introduction

In this article, we argue in favor of a formal model that views language as an integrated multimodal system, where syntax, prosody, and gestures are inherently linked. Our aim is to demonstrate that specific syntactic structures trigger a sensorimotor realization that encompasses not only intonation and phonology but also gestures. We argue that gestures, especially non-lexical co-speech gestures, are not merely an additional communicative

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channel; rather, they are an integral part of the grammatical system, generated and coordinated by the same syntactic principles that govern word order and sentence structure. In other words, they are a fundamental part of language.¹ The interesting peculiarity of the co-speech gestures examined here – as in previous work, see Giorgi and Dal Farra (2019), Petrocchi (2022), and Giorgi and Petrocchi (2025) – is that they do not add lexical content to the sentence, so it is not immediately clear why they appear systematically across speakers and languages. In what follows, we address some research questions aimed at clarifying this and related issues.

Specifically, we argue in favor of a representation of warning expressions that integrates both syntactic and pragmatic aspects. Along with discussing introspective data in Italian, we also provide experimental evidence aimed at clarifying the connection between syntax and sensorimotor implementation, especially prosody and gesture. Our primary focus is on the alignment of syntactic items with the gestural component, leaving the analysis of prosodic issues for future research. Our theoretical framework is the Minimalist theory, originally proposed by Chomsky (1995). In this work, we argue that warnings always involve an imperative – the warning call itself, which can be either silent or lexical – realized within the scope of an *apprehensive* projection. We present experimental evidence demonstrating that the apprehensive layer is always active, even when it is not explicitly realized through lexical items, instead manifesting through distinctive intonation and characteristic gestures.

This article is organized as follows: In section 2, we provide a brief introduction to the general framework we are working with. In section 3, we present our view regarding the syntax of warnings. In section 4, we present our research questions. In section 5, we provide a detailed description of the experiments performed, and in section 6, we present the statistical analyses. In section 7, we discuss the results from both quantitative and qualitative perspectives, and suggest some directions for future research.

2 Theoretical background

¹ The literature on gestures is very extensive, and it is impossible to present here an introduction to the whole field; however, see, among many others, McNeill (2005, 2000). Among the studies examining gestures in a formal perspective, the following primarily address the issue from a semantic point of view, e.g., Ebert & Ebert (2014), Schlenker (2018, to appear), Esipova (2019). Other studies also consider the relationship of gestures with syntax, see for instance Jouitteau (2004, 2007), Giorgi (2015, 2018), Giorgi & Dal Farra (2019), Ippolito (2021), Sailor & Colasanti (2020), Colasanti (2023), Ippolito et al. (2022), Giorgi & Petrocchi (2024, 2025).

A warning can be seen as a type of directive speech act, along with commands, orders, prohibitions, pleas, suggestions, and others. When using directives, the speaker aims to get the hearer to do something (Searle 1975). In other words, the main goal of a directive is to cause the addressee to perform an action. Warnings are linked to the typological category called *apprehensives*, which are linguistic tools that provide instructions to avoid an undesirable situation. Apprehensives are expressed in several languages through morphemes usually attached to the verb. In Italian, however, like in many other languages, there is no dedicated morphology, but instead, special constructions are used, as in the cases analyzed here.²

The warning expressions we will examine in this work are syntactically realized as presentative sentences, associated with a distinctive intonation and, as we will see, gestures. Consider, for instance, the following example:³

- (1) C'è un gradino!
There is a step!

This sentence, with the special intonation, is appropriately used when the speaker wants to warn the hearer about an immediate danger caused by a step. Unlike a simple statement about a nearby step, the sentence in (1) not only indicates that there is a step but also, in addition, alerts the listener that its presence could be dangerous.⁴

Our research question concerns the relationship between the syntactic representation of (1) and its function as a warning. One might argue, in fact, that the assertive sentence and the warning have the same representation and that the particular interpretation assigned to (1) is something added “later”, outside the domain of language grammar. Capitalizing on a proposal by Giorgi (2018, 2023), we will show that, on the contrary, syntax plays a very important role in

² For recent analyses of apprehensives across languages, see among the others Faller et al. (2025) and Dixon (2022).

³ We adopt here a generative theoretical perspective. A reviewer notes that generative grammar is traditionally not concerned with pragmatic issues. However, recently, several scholars have explored pragmatic problems within the generative framework – see for instance Domaneschi & Bambini (2021) – and gesture analysis in particular – see among many others, Schlenker (2018).

⁴ Therefore, we can say that, in this sense, a warning includes characteristics related to both an illocutive and a perlocutive speech act. See also Capone and Graci (2024) for the discussion of the notion of *pragmeme*, as a situated speech act that aims to modify situations and participants' roles.

identifying a warning as such, providing an explicit representation of the warning call, even when it is not overtly realized. In other words, the relevant information needed to interpret this sentence as a warning is represented in the syntax, allowing the sensorimotor apparatus and the conceptual system to correctly pair sound (and gesture) with meaning.

Let us point out, as discussed in Giorgi (2023), that in the generative framework, syntax has always been taken to play this kind of mediation, even if it was most often only implicitly assumed. Consider for instance the following string in Italian:

- (2) Ha telefonato
Lit: has called
‘he called’
‘did he call?’

This string can be associated with two different intonations. Prosodically, when it is a statement, it has a terminal fall, whereas when it is a polar question, it has a terminal rise.⁵ Therefore, one could theorize that the intonation disambiguates between the two possible meanings.

However, it has been argued since the very beginnings of generative grammar that interrogative sentences all have an operator in the syntactic representation responsible for the typical interpretation as questions. The operator is overtly realized in interrogative sentences as an interrogative phrase, introduced in English by a *wh*-word such as *who*, *what*, or by complex operators like *which boy* or *whose books*, and in Italian by interrogative such as *chi* (who), *che cosa* (what) or *quale ragazzo* (which boy). In *yes-no* questions the operator is not lexically realized. Consider the following structures, corresponding to the two interpretations of (2):⁶

- (3) [_T, pro [ha telefonato]]

⁵ For an analysis of intonation in Italian, see among the others Avesani (1990, 1995), D’Imperio (2002), Grice et al. (2005).

⁶ For earlier proposals on questions and interrogative operators, see Ross (1967) and Chomsky (1975). They accounted for these structures by means of transformational rules. Chomsky (1981) provided an analysis in the *Government and Binding* framework, dispensing with transformations and resorting to *principles*. Recent theories in the Minimalist framework from Chomsky (1995) on, do not significantly differ with respect to the hypothesis concerning the presence of the operator.

He called

(4) [_C, OP [_T, pro [ha telefonato]]

Did he call?

In (4), there is a silent operator, OP, in the left periphery, which has scope over the whole sentence.

Based on these observations, we can hypothesize that the difference in interpretation is not due to the prosodic realization but rather to the underlying syntactic structure, specifically the presence or absence of the silent, empty operator. In other words, the presence of the operator acts as the *trigger* for both a particular sensorimotor realization and the corresponding interpretation.

According to the Minimalist model, as originally proposed by Chomsky (1995) and further elaborated in subsequent works (Chomsky 2000, 2001, 2008, 2013), the syntactic representation is connected to the sensorimotor component on one side and to the conceptual one on the other. It can be represented as follows:⁷

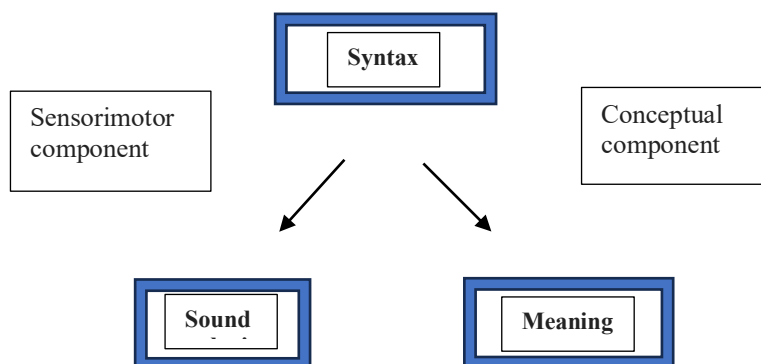


Figure 1: The minimalist model

As can be seen, in this model, there is no direct connection between speech and sound, which means that all correspondences between the two are mediated by syntax. A model that does not include a direct relationship between sound and meaning is less powerful than one that allows

⁷ The Minimalist framework in this case uses the term *syntax* to refer to *core syntax*, which is largely a synonym for *grammar*, intended as the competence of adult native speakers. In the text we are using *syntax* in the more traditional way, i.e., as identifying the structural representation of a sentence.

a dual relationship, i.e., one mediated by syntax plus a direct one. For this reason, it is to be preferred, being more constrained and therefore to be considered as the null hypothesis.⁸

3. The syntax of warnings

In this section we are going to briefly consider the syntax of warnings and the implications on our research questions.

In example (1), we gave an example of warning. We repeat it here for simplicity, with the relevant context:

(5) Context: I am walking with my friend, and I see that he is going to trip over a step and might fall. I warn him and say:

(6) C'è un gradino!

There is a step!

There are also other possible ways in which the speaker might convey such a warning:

(7) (Sta') attento, c'è un gradino!

Lit: (you stay) attentive, there is a step

(8) (Sta') attento che c'è un gradino!

Lit: (you stay) attentive that there is a step

'Watch out that there is a step!'

In example (7), the warning call is followed by the presentative sentence, in (8) the presentative sentence is introduced by the complementizer *che* (that).⁹ The warning call can optionally include the imperative form of the auxiliary verb *stare* (stay) followed by the warning adjective *attento*.¹⁰

⁸ For further discussion of this and related points, see Giorgi (2023).

⁹ On the syntax of presentative sentences, see among the others Moro (2017).

¹⁰ As an auxiliary, *stare* (stay) appears in other verbal locutions, such as for instance the progressive periphrasis:

i. Gianni sta mangiando un panino
Lit. Gianni stays eating a sandwich
'Gianni is eating a sandwich'.

We are not going to pursue here the issues related to the distributions of the various auxiliaries in Italian and refer the reader on literature on the topic, see among the others Bertinetto (1991).

Moreover, in these contexts, the speaker might also utter the warning call without mentioning the nature of the danger:

- (9) (Sta') attento!
(you stay) attentive!
'watch out!'

The warning call can also be realized as a nominal form, such as *attenzione* (attention), optionally preceded by the imperative of the light verb *fare* (do), with or without *che* (that), as in the following examples: ¹¹

- (10) (Fa') attenzione, c'è un gradino!
Lit: (you do) attention, there is a step
(11) (Fa') attenzione che c'è un gradino!
Lit: (you do) attention that there is a step
'Pay attention that there is a step!'
(12) (Fa') attenzione!
Lit: (you do) attention!
'Pay attention!'

¹¹ In these examples, we have glossed *che* as *that*. However, it seems to us that these examples cannot be interpreted as featuring subordinate clauses, since *attento* (attentive) and *attenzione* (attention) do not unequivocally admit of a clausal complementation and are compatible with the intervention of the preposition *a* (to):

- (i) ??Gianni è attento che Maria mangi
Lit: Gianni is attentive that Maria eats
(ii) Gianni è attento **a** che Maria mangi
Lit: Gianni is attentive **to** that Maria eats
'Gianni watches out that Maria eats'
(iii) ??Gianni fa attenzione che Maria mangi
Lit: Gianni does attention that Maria mangi
(iv) Gianni fa attenzione **a** che Maria mangi
Lit: Gianni does attention **to** that Maria eats
'Gianni watches out that Maria eats'

The sentences with *a che* (to that), though belonging to a somewhat high register, are perfectly grammatical, whereas the prepositionless equivalent is degraded.

The presentative clause, when realized, is better characterized as the *reason why* the speaker is uttering the warning call rather than as a *complement* to the warning call. It seems to us that the *che* appearing in these cases could also be written as '*ché*,' a somewhat formal lexical item short for *perché* (why, because).¹²

Our syntactic proposal for these structures is as follows: the proposition indicating the origin of the danger – in this case, *c'è un gradino* (there is a step) – is embedded within a larger structure, which can also include a warning call. This structure is a *micro-discourse* (Giorgi 2023), i.e., a structure resembling a single sentence but formed by two separate sentences, each expressing a speech act, connected by a *discourse head*. The warning call, in fact, can also appear alone, without any other linguistic context, as shown in examples (9) and (12) above. The two components of these constructions – the warning call and the clause indicating the source of danger – can therefore be viewed as distinct. Both parts are characterized by specific intonation and gestures.

Concluding, we argue that this sequence consists of two independent sentences linked by a silent discourse head. When present, the lexical item *che* indicates a causal relationship and appears in the left periphery of the presentative sentence, within a dedicated projection. When *che* is absent, the causal relationship is unspoken and is inferred from the context. The structure can be represented as follows:¹³

(13) [[*warning call*] DIS [(*che*) *description of danger*]]

As proposed above, these structures are micro-discourses, i.e., sequences of two or more full-

¹² Rizzi (2001) locates the interrogative *perché* in a projection dubbed INT(errogative)-P, appearing in the left periphery lower than the high complementizer Force. One might wonder whether the *perché* expressing a causal value – *because* – is the same as the *perché* expressing an interrogative one – *why*. At this stage of our research, we do not have an answer to this particular question, but we believe that even if the two *perché* might not be located in the same projection, they nonetheless occupy a position in the left periphery, lower than Force-P. The issue is, however, quite complex, and we are not going to pursue it further here.

¹³ For a discussion of micro-discourses in expressive language, such as surprise and disapproval contexts, see Giorgi (2018). For a brief discussion of Hanging Topics and vocatives as well, see Giorgi (2023). Note also that *che* is admitted only if the warning call is overtly realized. A reviewer suggests that *che* might be ambiguously characterized as a discourse head as well. Given the causal interpretation, however, we are inclined to consider it a variant of *perché* (because), and therefore a head belonging to the left periphery. The matter however requires further investigation.

fledged sentences that, at first glance, appear to be a single unit. However, upon closer scrutiny, it turns out they are not, as they constitute a hierarchically ordered sequence of independent sentential items, which is in fact the characteristic of discourses.

The structure in (13), although not fully detailed, highlights the main points of our argument. While we won't conduct an in-depth syntactic analysis of the warning call on one side and the presentative sentence on the other, it is important to note that a warning call in Italian usually appears as an imperative sentence. The forms *sta'* (stay) and *fa'* (do), which typically precede *attento* and *attenzione* respectively, are imperative verbal forms. Imperatives instantiate a peculiar structure, activating the left periphery of the sentence; for a detailed analysis of imperatives, see, among others, Rupp (2003), Alcázar & Saltarelli (2014), Zanuttini (2008), and Zanuttini et al. (2012). Our goal here is not to discuss or compare their proposals, since our main focus in the following sections is to demonstrate that the presence or absence of lexical content to express the warning call – i.e., of the imperative – in a certain sense does not matter, because intonation and gesture provide the necessary cues for the correct interpretation. Our basic tenet is that there is no direct link between the sensorimotor system and the conceptual system, so the relevant information must come from the syntax, as intonation and gesture are realizations of syntactic information.

4 The hypothesis: silent imperatives and language multimodality

Summarizing so far: *c'è un gradino!* (there is a step!) is understood as a warning because of intonation and gesture, even when the warning call, for example, *sta' attento!*, is not being uttered.

Our hypothesis is that the warning call is always present in the syntax, even when it is not uttered and that its presence in the representation turns a presentative sentence into a warning. The silent imperative, therefore, acts as the trigger for the characteristic intonation and gesture associated with this structure.¹⁴

Note that *attento* and *attenzione*, by themselves, are not warning calls, i.e., they are not lexically specified as such, in that they can also appear in neutral contexts, as the following ones:

¹⁴ A reviewer points out that there are many other types of warnings. This is an important observation. Here we only consider warnings including an explicit or implicit imperative warning call. It is certainly crucial for our future investigation to consider how broad our proposal can be and how far it can be extended to other types of warnings.

(14) Gianni è attento ad ogni dettaglio

Lit: Gianni is attentive to every detail

(15) Gianni fa attenzione ad ogni dettaglio

Lit: Gianni makes attention to every detail

‘Gianni pays attention to every detail’

Therefore, we propose that the warning call – i.e., the one with the imperative – is in the scope of an *apprehensive* projection.¹⁵ Thus, the structure in (13) can be made more precise as follows:

(16) [[Appr-P APPR₀ [*imperative*]] DIS₀ [(*che*) *description of danger*]]

The apprehensive head does not correspond to a lexical entry but is read off the interface with the sensorimotor apparatus and realized as intonation and gesture. According to this hypothesis, therefore, there shouldn’t be any difference between the sentences with an explicit warning call and those without.

In this article, we focus on the gestural pattern and formulated the following operational research questions: a) Is there a gestural pattern characteristic of warnings in particular? b) What happens when an explicit warning call precedes the sentence? I.e., are there statistically significant differences in gesture production – quantitatively and qualitatively – between warnings preceded by warning calls and warnings without warning calls? c) Are there statistically significant differences between the gesture production in warnings, with and without warning calls, and non-warning sentences, in particular simple affirmative and simple interrogative sentences?

¹⁵ We refer to this syntactic node apprehensive projection – Appr-P – in analogy to the apprehensive phenomena mentioned in section 1. However, because of space limits, we will not explore possible connections between the apprehensive verbal morphemes in different languages and the Italian phenomena discussed in this article. It’s important to note, though, that in some languages, a category may show up as a morpheme, while in others, it appears as an adverb or a specific phrase. A similar relationship between a high, left-peripheral projection and a morphological form on the verb has been studied in Palermitan, a Sicilian dialect (Giorgi and Sorrisi, 2018). In their study, the authors show that the morphological realization of an evaluative morpheme on the verb is syntactically connected to the high evaluative projection Eval-P.

For our hypothesis to hold, the answer should be “yes” to question (a), “no” to question (b), and “yes” to the double question in (c).

5 Methods and materials

We conducted two experiments with warning contexts plus a control condition, involving 51 participants in total, across three different groups. The first experiment—Experiment A, with 23 participants—focused on identifying a gestural pattern for warnings. The second experiment—Experiment B, with 13 participants—investigated potential differences in gestural patterns between sentences with an explicit warning call and those without. Lastly, we included a control group of 15 participants that produced only non-warnings, such as affirmative and interrogative sentences.

Experiment B repeated Experiment A by adding 6 stimuli to the ones already present in A. The total number of subjects, responding to the 8 stimuli present both in A and B, was therefore 36, with 13 of them responding to a longer list of stimuli, i.e. 14. Next, we will describe the methods and materials for each experiment.

5.1 Experiment A

5.1.1 Participants

23 native Italian speakers participated in the experiment (21 females, 2 males; average age = 31 years). All completed a questionnaire to evaluate their linguistic background. Our sample size was determined by an a priori power analysis to ensure adequate statistical power to detect the hypothesized effects.¹⁶

5.1.2 Materials

We used a Sentence Repetition Task (SRT) because it proved to be a reliable method that can easily be adapted to our experimental needs.¹⁷ We created eight specific contexts designed to

¹⁶ An a priori power analysis, performed with the R package *simr*, showed that the proposed experimental design is well-powered to detect the target effect. Based on an alpha level of 0.05, a desired power of 80%, and an expected medium effect size ($IRR = 1.5$), our analysis indicated that a sample of 23 participants for Experiment A is sufficient, achieving a statistical power of 99.00%.

¹⁷ In spoken language research, SRT has mainly been used to study language acquisition and development in children. It has also proven to be a reliable tool for clinical assessment, providing insights into children’s sentence-level abilities and serving as a clinical marker for specific language impairment (Polišenská & Roy, 2014). SRTs have consistently shown high

elicit warning utterances. Each context followed the same structure: after presenting a conversational scenario involving two friends – one of whom is the participant – a potentially dangerous situation was introduced. The participant was then asked to read and repeat, after the screen went blank, an utterance intended to warn their friend about the imminent danger. All target sentences are of the warning presentative type (see section 4). Examples of the contexts and their corresponding target sentences are provided below.

- (17) Context, read aloud in Italian by the experimenter: “You and your friend are driving around, and your friend is the one behind the wheel. You notice that the traffic light is red, but your friend is not slowing down. You warn him and say”:

Sentence to be repeated by the subject: “*È rosso*” (it is red!)

- (18) Context, read aloud in Italian by the experimenter: “You and your friend are eating soup. You notice that there is a fly in your friend’s bowl. You warn him and say”:

Sentence to be repeated by the subject: “*C’è una mosca nel piatto*” (there is a fly in the soup!).

The warning scenarios were designed to cover a wide range of realistic risk situations, from high-stakes events (e.g., running a red light at an intersection) to low-stakes events (e.g., finding a fly in food). This approach allowed us to control for the role of the warning itself, regardless of the risk’s severity. The instructions given to the subjects did not include any references to prosodic or gestural cues. Specifically, no mention of prosody or gestures was included in either the auditory (read aloud) contexts or the task instructions. The sentences to be repeated were shown in written form on a computer screen without punctuation. These methodological choices aimed to eliminate possible prosodic or gestural priming and prevent participants from getting communicative hints from the materials themselves. Controlling for

internal consistency and strong correlations with independent measures of language proficiency (Rujas et al., 2021). Importantly, these results have been replicated in sign language studies, confirming the tasks cross-modal applicability. See, Schönström and Hauser (2021), Bogliotti et al. (2020), Schönström et al. (2022). While much of the literature validating the SRT comes from studies on language acquisition and development – both in spoken and signed language – its methodological strengths make it an appealing tool for a broader range of linguistic investigations.

these factors is essential in multimodal research, as both prosody and gestures can be affected by subtle input cues, such as visual or typographic markers and vocal emphasis in model utterances. By removing these cues, we made sure that any prosodic or gestural features produced during the task truly reflected participants' spontaneous planning and expression. This not only improves the ecological validity of the data but also strengthens the causal link between the communicative context and the observed multimodal behaviors, allowing for more accurate interpretation of the results.

When we examined the results of the first experiment, we noticed that several subjects spontaneously added an explicit warning call to the sentence they were asked to repeat. For example, instead of repeating *è rosso* (it is red), they added *sta' attento* (watch out), which was not part of the stimulus, responding with a more complex sentence: *sta' attento! È rosso!* (watch out! It is red!).

We decided, therefore, to explicitly examine this issue and designed experiment B with 6 additional stimuli, including a warning call, for a total of 14 stimuli, as we will illustrate in section 5.2.

5.1.3 Procedure

Participants were instructed to stand in a relaxed posture, with their arms naturally resting at their sides about 40 cm from the computer screen. To promote comfort and a natural demeanor in front of the camera, the experiment began with a background questionnaire designed to gather information about the participant's linguistic history. This phase was conducted with the camera already turned on and served both practical and acclimation purposes. The actual task only started after a brief conversation between the experimenter and the participant, allowing the latter time to settle into the experimental setting.

After listening to the context – read aloud by the experimenter – the participant was shown a target sentence to repeat. Stimulus presentation was controlled by a PC running PowerPoint. The sentence appeared on the screen for a few seconds, during which the participant read it silently. Once the stimulus disappeared and a blank slide appeared – as a visual cue – the participant was expected to repeat the sentence aloud from memory, imagining themselves in the previously described scenario. The instruction was to repeat the sentence as naturally as possible, imagining themselves in the described situation and producing the utterance as if it were part of a real-life interaction.

5.1.4 Data Transcription and Analysis

The participants were videotaped, and the resulting footage was analyzed using ELAN.

The test was originally recorded without interruptions. The video was segmented using Kdenlive 23.04.1, and only the segments where the speaker repeated the target sentence were extracted. As a result, each participant had 8 clips and 8 audio files, each showing one of the 8 target stimuli, totaling 184 stimuli (8 stimuli $\times$ 23 participants).

The video data were transcribed in ELAN version 6.9. To ensure accurate timing with co-speech gestures, the gestures were annotated and synchronized with the video. For this purpose, a dedicated ELAN template was created with the following annotation tiers: manual gestures (handshape), head movements, eyebrow movements, eye aperture, torso movements, speech transcription, and additional notes. The main goal of the ELAN analysis was to identify both manual and non-manual gestures and to investigate their potential alignment with the linguistic material.

The audio recordings were transcribed using Praat version 6.4.23. Transcriptions were time-aligned with the audio signal and saved in TextGrid format for further analysis. The analysis of the intonational properties, however, is deferred to further investigation.

5.1.5 Gesture Coding

We coded the gestures produced by participants while they repeated the target stimuli.

Regarding manual gestures, our analysis centered on the stroke phase of each gesture, which we consider the most important part of the movement. Following Kendon's (2004) framework, we identified and coded the stroke within each gesture unit, paying close attention to handshape configuration, as it holds key semantic or pragmatic information. Kendon (2004) describes gestures as comprising several distinct phases. The Preparation Phase occurs when the hands or body get ready for the gesture, usually before any major movement happens. The Stroke Phase is the crucial moment when the main movement occurs. The Hold Phase refers to the period during which the gesture is kept in a specific position, maintaining the stroke configuration. The Retraction Phase involves withdrawing the gesture or returning to a neutral position. These phases together form the "gesture unit," capturing the full range of movement and conveying meaningful action in communication. Kendon's framework remains influential in gesture research. Much work has focused on analyzing the Stroke Phase (Sowa, 2006), examining its timing with speech—especially the alignment between the stroke and prosodic

features like pitch accents.¹⁸ Additional studies show that gestures can "wait" for speech elements, with the stroke potentially being "suspended" (Seyfeddinipur, 2006) to synchronize with verbal components. Recent research, which revisits gesture phases with a more detailed and gesture-centric approach, continues to build on Kendon's original division of phases to describe the specific features and internal distinctions that separate each phase (Bressem & Ladewig, 2011). For non-manual gestures, we coded the stroke as the first complete occurrence of the gesture itself (Church et al., 2017), which includes full eye opening, head tilt, and forward torso movement. We also coded the stroke along with its hold phase, the period during which the gesture is maintained beyond the stroke and before retraction begins. The entire gesture unit (preparation phase, stroke, and hold phase) was considered to represent the full scope of the gesture within the sentence

5.2 Experiment B

Experiment B used the same procedures and methods for data transcription and analysis as Experiment A to ensure comparability and allow parallel result analysis. However, Experiment B used a larger set of stimuli.

5.2.1 Participants

13 native Italian speakers took part in the experiment (6 females, 7 males; average age = 30 years). All participants filled out a questionnaire to assess their linguistic background, aiming to identify potential influences from dialect use and education level. Our sample size was determined by an a priori power analysis to ensure enough statistical power to detect the expected effects.¹⁹

5.2.2 Materials

Consistent with our introspective observations (see sections 4 and 5.1), the first experiment revealed that several participants naturally prefaced the presentative warning sentence with a warning call (e.g., *Attento*, literally: "Watch out") during repetition. To further explore this

¹⁸ See Nobe (2000), Loehr (2006 2012), Queck et al. (2002), Yassinik et al. (2004), Türk & Calhoun (2023).

¹⁹ An a priori power analysis conducted using the *simr* package in R revealed that our proposed experimental design has a high probability of detecting the target effect. With a specified alpha level of 0.05, a desired power of 80%, and an expected medium effect size (IRR=1.5) our power curve analysis showed that a sample of 13 participants for Experiment B is more than adequate, achieving a statistical power of 91,20%.

phenomenon and carefully examine potential differences in gesture production between warnings preceded by warning calls and those without, a new set of stimuli was created and tested on a new controlled group of participants. Therefore, this second stimulus set for the SRT included the 8 stimuli from the previous experiment plus 6 additional stimuli, totaling 14. In total, 182 stimuli were presented ($14 \text{ stimuli} \times 13 \text{ participants}$). Specifically, compared to Experiment A, Experiment B also included 6 sentences with an explicit warning call, as the following ones: *attento* (watch out), *attento è rosso* (watch out it is red), *ehi attento c'è un bambino* (Hey watch out there is a kid).

5.3 Control group C

Experiment C followed the same procedure and methodology for data transcription and analysis as Experiments A and B. However, Experiment C utilized a completely different set of stimuli for the SRT.

5.3.1 Participants

15 native Italian speakers participated in the experiment (8 females, 7 males; average age = 37 years). All participants filled out a questionnaire to assess their linguistic background, aimed at identifying potential influences of dialect use and educational level. Our sample size was determined through an a priori power analysis to ensure enough statistical power to detect the hypothesized effects.²⁰

5.3.2 Materials

The group of participants in Experiment C repeated two sets of control stimuli, consisting of neutral affirmative and interrogative sentences, free of any warning nuance. The first stimulus set included 7 canonical open questions, which were balanced regarding the *wh*-phrases used and matched in lexical and phonological aspects. The second set contained 7 neutral affirmative sentences; this set was balanced for lexical and phonological load as well. As a total, we had 210 stimuli ($14 \text{ stimuli} \times 15 \text{ participants}$). Below are two examples from the control stimulus

²⁰ An a priori power analysis, conducted using the *simr* package in R, demonstrated that the proposed experimental design is well-powered to detect the hypothesized effect. Based on a specified alpha level of 0.05, a desired power of 80%, and an expected medium effect size (IRR=1.5), our analysis indicated that a sample of 15 participants for Experiment C is more than adequate, achieving a statistical power of 95.60%.

battery:

(19) Context (read aloud in Italian by the experimenter): “You know that Maria went shopping at the new mall. Curious, you ask:”

Sentence to be repeated by the participant: “*Cosa ha comprato Maria?*” (What did Maria buy?)

(20) Context (read aloud in Italian by the experimenter): “A friend of yours is planning an outdoor outing for tomorrow. You inform her by saying:”

Sentence to be repeated by the participant: “*Domani è prevista pioggia*” (Tomorrow it’s going to rain.)

We followed the same procedure and methodology used in Experiments A and B to ensure that the results could be reliably compared with those of the previous studies. The main goal was to compare the multimodal realization of neutral interrogative and affirmative sentences with warning structures, to identify any distinctive features specific to warning sentence production. In particular, we wanted to examine eyebrow movements, since they are well-documented as essential to question production in many sign languages. As will become clear in the next section, in fact, eyebrows play a crucial role in the multimodal realization of warning expressions.²¹

6 Results

All statistical analyses were conducted in R (version 4.5.1). Mixed-effects models were fitted using the *lme4* package (version 1.1-35), with the function *glmer* applied to account for both participant and item variability. Post-hoc pairwise comparisons were performed using the *emmeans* package (version 1.10.3).

6.1 Experiment A

To investigate whether the overall frequency of gestures was modulated by the type of linguistic stimulus, we fitted a Generalized Linear Mixed Model (GLMM) with a negative binomial distribution. The dependent variable was the number of gestures produced per trial

²¹ On the role of eyebrow movements, see Pfau et al. (2012) for sign languages. See also Nota et al. (2023) and see Ekman (1979, 1993).

(Gesture_n), and the main predictor was the stimulus (Stimulus1–Stimulus8). Participant ID was included as a random intercept to account for individual variability. The model revealed a significant random effect of participant (Variance = 0.655, SD = 0.809), indicating substantial inter-individual variation in gesture frequency. No overdispersion was detected (dispersion ratio = 0.819, $p = 0.52$), indicating an appropriate model fit.

Regarding fixed effects, none of the stimuli produced a statistically significant difference in the number of gestures. Although the estimates suggested a numerically higher number of gestures for Stimuli 3 and 4, and a lower number for Stimulus 7, these effects did not reach conventional levels of statistical significance ($p > .05$). Notably, Stimulus 7 approached marginal significance (*Estimate* = -0.693 , $p = 0.073$), indicating a potential trend toward reduced gesture production in response to lower-warning stimuli.

Post-hoc pairwise comparisons confirmed the absence of statistically significant differences between stimulus conditions, after Tukey adjustment for multiple comparisons. This suggests that, overall, gesture frequency does not robustly vary as a function of the linguistic stimuli used to elicit warning-related utterances.

These findings support the hypothesis that the gestures observed reflect a broader pattern associated with the pragmatic structure of these specifically designed utterance types, which behaved consistently in eliciting the targeted pragmatic nuance. The gestures observed include Raised Eyebrows (E), Body Forward (B), Head forward (H), Furrowed Eyebrows (F), Widened Eyes (W) and Nod (N):

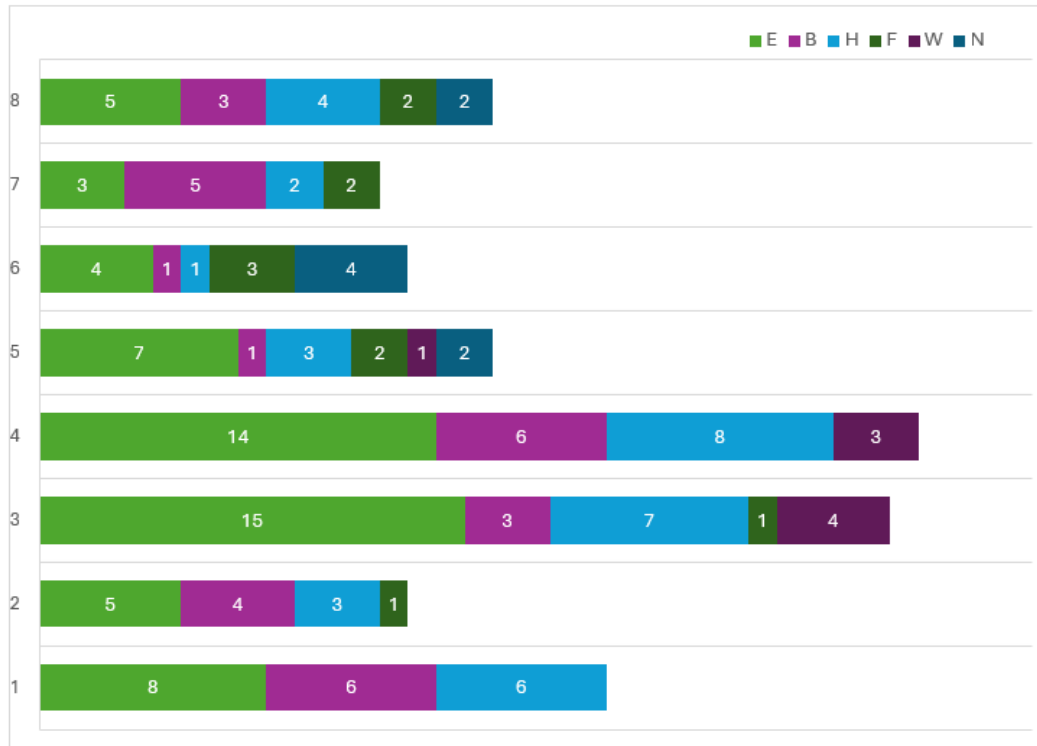


Figure 2: Distribution of gesture types across stimuli

In Figure 2, each bar represents the total number of gestures produced per stimulus, with individual segments indicating specific gesture types. The y-axis displays the total number of gestures, while the x-axis labels the stimuli from 1 to 8. Furrowed Eyebrows (F), Widened Eyes (W), and Nod (N) occurred too infrequently to reach statistical significance. In contrast, gesture types E (Raised Eyebrows), H (Head Forward), and B (Body Forward) were consistently observed across all stimuli. When more than one of these gestures occurred, they were always performed simultaneously. Their presence in every stimulus, even if not realized by each participant, suggests a potential pattern of co-occurrence or functional complementarity, as shown in Figure 3.

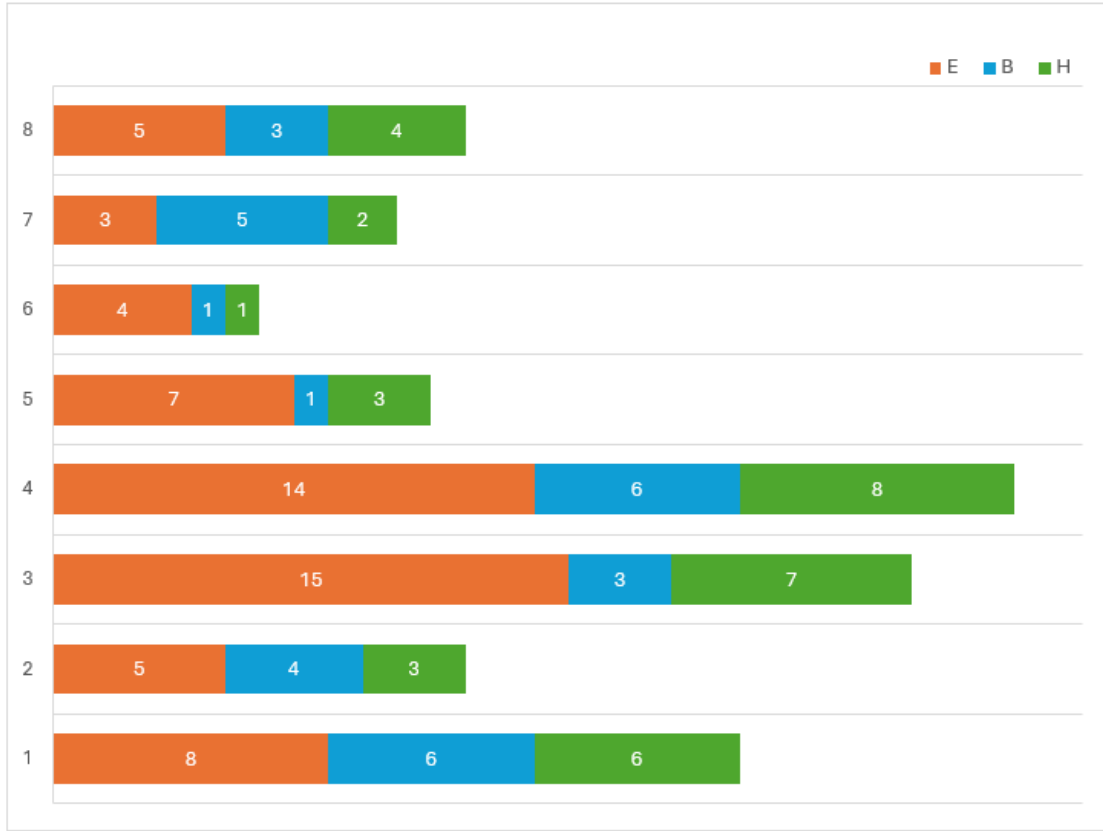


Figure 3: Distribution of Gesture Types E, B and H across Stimuli

In further analysis, we examined the effect of the independent variable *Stimulus* on the likelihood of producing specific gesture types, focusing on the three most frequently observed gestures across conditions: Raised Eyebrows (E), Body Forward (B), and Head Forward (H). Separate mixed-effects logistic regression models were fitted for each gesture, with *Stimulus* as a fixed effect and *Participant* (ID) as a random intercept.

6.1.1 Gesture E (Raised Eyebrows)

Random effects indicate substantial individual variability (Variance = 2.876, SD = 1.696). Fixed effects reveal a significant influence of the stimulus on the production of Gesture E. In particular, the probability of producing the gesture increases significantly for Stimulus 3 (*Estimate* = 1.89, *p* = .015) and Stimulus 4 (*Estimate* = 1.61, *p* = .035) and decreases significantly for Stimulus 7 (*Estimate* = -1.84, *p* = .048). Other stimuli show no significant effects. These findings align with the hypothesis that Gesture E systematically marks warning-related content. Notably, Stimuli 3 and 4 were designed to express high-level danger (i.e., crossing at a red light; a child potentially being hit by a car), whereas Stimulus 7 encoded

minimal danger (i.e., pasta water about to boil over). Thus, gesture E appears to be highly sensitive to the perceived degree of danger and is systematically associated with warning constructions.

6.1.2 Gesture B (Body Forward)

Gesture B shows very high inter-individual variability (Variance = 8.048, SD = 2.837), suggesting a gesture more subject to personal inhibition. No significant effects are found for Stimuli 2, 3, 4, 7, or 8. However, a significant reduction in the probability of occurrence is observed for Stimulus 5 (*Estimate* = -3.50, *p* = .017) and Stimulus 6 (*Estimate* = -3.50, *p* = .017). Notably, Stimuli 5 and 6 express minimal levels of danger (i.e., *there is mold on your cheese* and *a storm is approaching*).

Thus, Gesture B appears to be sensitive to the perceived degree of danger and systematically associated with warning constructions. However, it is generally present in medium- to high-danger contexts, and it tends to disappear when the perceived danger is low (e.g., mould, storm). Thus, gesture B behaves similarly to Gesture E in functioning as a warning gesture, but it appears to be more voluntary and more easily inhibited.

6.1.2 Gesture H (Head Forward)

Random effects again indicate significant inter-individual variation (Variance = 2.694, SD = 1.641). However, none of the stimuli reached statistical significance in predicting Gesture H production. Only Stimulus 7 shows a marginal trend toward reduction (*Estimate* = -1.82, *p* = .084). Gesture H does not appear to be modulated by the gradient level of the warning and its presence remained consistent across all experimental conditions.

6.1.3 Overall considerations

An important insight from the data relates to individual differences. Specifically, this variability involves how often gestures are made, not which types are used. All participants who responded to warning stimuli with gestures consistently used the same types – E, H, and B – though the frequency varied. This shows that inter-individual differences are in the rate of gesture production rather than the choice of gestures, and this is crucial for our study. Some participants gestured much less than others, but when they did gesture, they always used the

same types.²² Looking at the data, 100% of participants who gestured used Raised Eyebrows (E); 85% used Head Forward (H), and 55% used Body Forward (B). Additionally, among the 20 of 23 participants who gestured, 70% employed at least two components of the E-H-B warning pattern. Overall, out of 160 gestures recorded, the E-H-B pattern made up 53.12% of all cases.

Finally, although other gesture types were also observed – namely F (Furrowed Eyebrows), S (Squinted Eyes), Sh (Head Shake), N (Nod), D (Disgust), U (Head Upward), and W (Widened Eyes) – their occurrence was limited and it did not reach statistical significance. In particular, S, Sh, D, and U occurred in only a few instances and were produced in fewer than three stimuli, with considerable variability across speakers; for this reason, they were excluded from the analysis. The disgust (D) gesture was explicitly limited to low-warning contexts related to the risk of eating a soup with a fly or cheese with mold.

By contrast, Furrowed Eyebrows (F), Widened Eyes (W), and Nod (N) occurred in three stimuli or more and were therefore retained in the graph in Figure 2. However, their frequency was too low to reach statistical significance, and they could not be modeled in the R analysis.

6.1.4 Statistical Analysis of Gesture Co-occurrence (E-H-B pattern)

To investigate whether gestures E (Raised Eyebrows), H (Head Forward), and B (Body Forward) constitute a coordinated pattern, we performed a log-linear analysis on a three-way contingency table. The dependent variables were the binary presence or absence of each gesture (E, H, and B), and the model was used to test for significant interactions. Initially, we tested for a three-way interaction, which was found to be non-significant ($p=0.52$). This result indicates that the three gestures do not form a single, inseparable pattern. Following this, we proceeded to test for two-way interactions. The analysis revealed a highly significant interaction between Gestures E and H ($p<0.001$), as well as a highly significant interaction between Gestures H and B ($p<0.01$). Conversely, the interaction between Gestures E and B was not statistically significant ($p=0.43$). These findings suggest that while the three gestures do not co-occur as a unified triad, they form statistically robust co-occurrence patterns in pairs, with the Head Forward gesture (H) acting as a crucial component that links the other two gestures.

²² One of the important issues is presumably the fact that a recorded experimental session is by itself an inhibitory factor. On the other hand, purely ecological data are difficult to analyze in that they are not systematic with respect to the variables involved.

6.2 Experiment B

The same analysis fitting a Generalized Linear Mixed Model (GLMM) with a negative binomial distribution was conducted in this second experiment. The dependent variable was the number of gestures produced per trial (Gesture_n), and the main predictor was the stimulus (Stimulus1–Stimulus14). Participant ID was included as a random intercept to account for individual variability.

The model revealed a significant random effect of participant (Variance = 1.902, SD = 1.379), confirming substantial inter-individual variation in the frequency of gestures produced. No overdispersion was detected (dispersion ratio = 0.186, $p = 0.432$), indicating an appropriate model fit.

Regarding fixed effects, all p -values for the individual stimuli were well above the conventional significance level of 0.05 (e.g., Stimulus2, $p = 0.244$; Stimulus3, $p = 0.819$). Thus, the analysis found no sufficient evidence to suggest that the type of stimulus significantly influences the number of gestures produced. Instead, the results indicate that gesture production may reflect a more general pattern associated with the pragmatic structure of the utterance type used, rather than being modulated by specific stimulus characteristics.

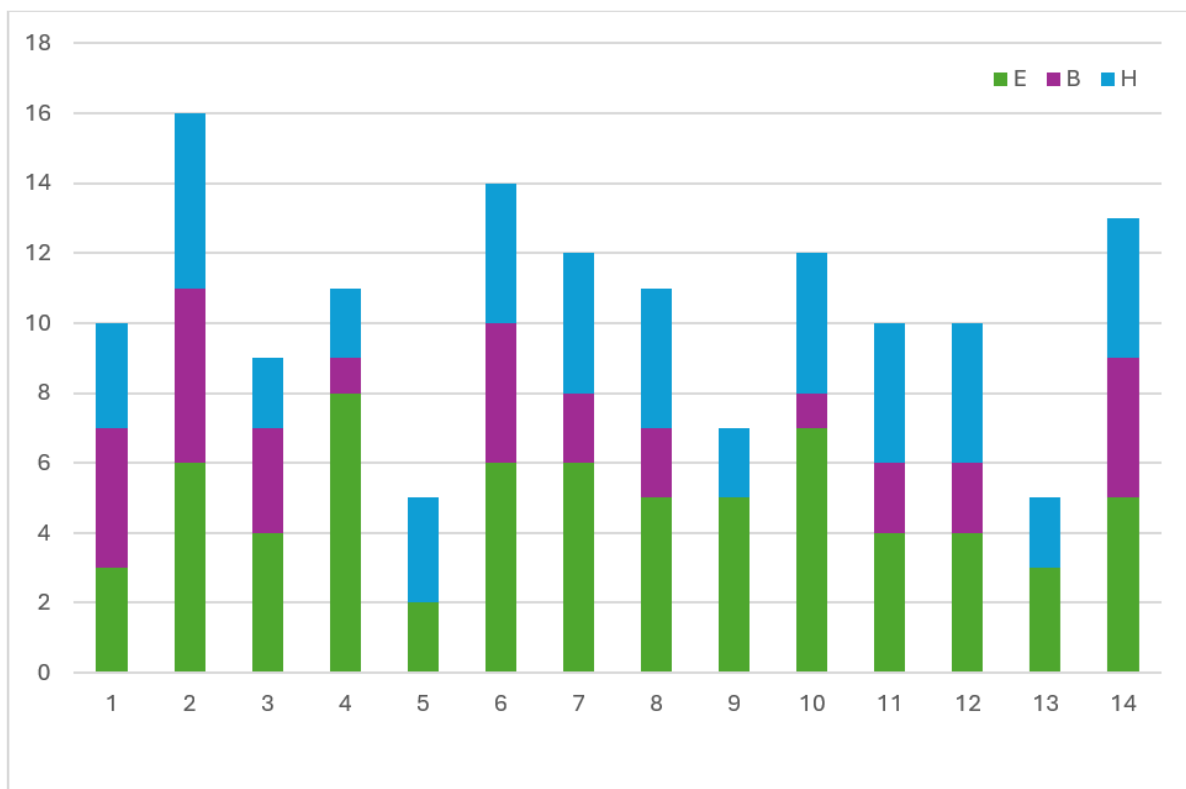


Figure 4: Distribution of Gesture Types E, B and H across Stimuli

The graph in 4 reveals that gestures E (Raised Eyebrows), H (Head Forward), and B (Body Forward) consistently co-occur across different stimuli, forming a cohesive and regular pattern. In this case as well, the three gestures were invariably realized simultaneously whenever they occurred. This second experiment also included a range of stimuli from high- to low-warning conditions. Notably, in the low-warning stimuli – Stimulus 5 (a fly on a plate), Stimulus 9 (the risk of an approaching thunderstorm), and Stimulus 13 (the pasta water about to boil over) – gesture B is absent. An additional gesture, Disgust (D), was also observed; however, its distribution proved inconsistent and far from statistical significance. In particular, it appeared only in Stimulus 5 (a fly on a plate) and Stimulus 11 (mold on cheese).

In a second analysis, we investigated the effect of the independent variable *Stimulus* on the likelihood of producing specific gesture types, focusing on the three gestures most observed across conditions: Raised Eyebrows (E), Body Forward (B), and Head Forward (H). Separate mixed-effects logistic regression models were fitted for each gesture, treating *Stimulus* as a fixed effect and *Participant* (ID) as a random intercept.

6.2.1 Gesture E (Raised Eyebrows)

The model used was a Generalized Linear Mixed Model (glmer) with a binomial distribution, treating *Stimulus* as a fixed effect and Participant ID as a random intercept. The initial model returned a convergence warning, prompting an optimization with the *bobyqa* optimizer. This optimization successfully resolved the convergence issue, and a comparison of the model summaries confirmed that the parameter estimates and p-values remained virtually unchanged, validating the original findings.

The model revealed substantial inter-individual variability in gesture frequency, with a significant random effect of participant (Variance = 11.3, SD = 3.361). Regarding fixed effects, the stimulus significantly influenced the probability of producing Gesture E. Specifically, the probability of producing the gesture increased significantly for Stimulus 4 ($p < .01$), Stimulus 7 ($p < .05$), and Stimulus 10 ($p < .05$). While not reaching conventional levels of statistical significance, Stimulus 2 and Stimulus 6 also showed a marginal trend ($p < .1$). All other stimuli did not show a significant effect. Interestingly, Stimuli 4, 7, 10, 2, and 6 all represented a high-warning level, each depicting variations of a context involving the risk of running a red light and hitting a child crossing the street. Notably, this pattern emerged despite differences in the presence of explicit warning calls: Stimulus 7 (a presentative warning sentence) lacked an

explicit warning call, Stimulus 10 included the warning call alone (*Watch out*), and Stimulus 4 combined the warning call with the presentative warning sentence (*Watch out, it is red*). Similarly, Stimulus 2 featured the warning call alone (*Watch out*), whereas Stimulus 6 included the presentative warning sentence (*There is a child*).

These findings suggest that, from a statistical point of view, no differences are observed in the realization of these apparently different sentence types, at least as far as Gesture E is concerned. Moreover, these results replicate those of Experiment A, reinforcing the conclusion that Gesture E is a core element of the multimodal warning pattern, independently of the presence or absence of an explicit warning call.

6.2.2 Gesture B (Body Forward)

Our initial analysis of Gesture B revealed a significant statistical issue related to data sparsity. Specifically, the gesture was never observed for Stimuli 5, 9, and 13, leading to a phenomenon known as complete separation. This issue, which persisted despite our attempts to optimize the model, occurs when a predictor variable perfectly predicts the outcome, causing the maximum likelihood estimates to become infinite. Consequently, the model for Gesture B was unable to converge properly. Thus, to ensure a robust and valid analysis, we chose to exclude these three stimuli from the logistic regression model. This decision is further justified by the fact that Stimuli 5, 9, and 13 all represent low-warning conditions (i.e., a fly on a plate, an approaching storm, and pasta water boiling over). This approach allows us to focus on contexts where the gesture was actually produced, enabling a more stable model fit and providing meaningful insights into the factors that modulate the frequency of this specific gesture.

After filtering the data, a re-analysis using a mixed-effects logistic regression for Gesture B (Body Forward) revealed no statistically significant fixed effects for any of the remaining stimuli. A substantial random effect of participant was observed (Variance = 2.77, SD = 1.66), indicating considerable inter-individual variability in gesture frequency. The model showed that the frequency of Gesture B was not significantly influenced by the high-warning stimulus content. Thus, in this case as well, the presence or absence of the warning call does not affect the occurrence of Gesture B in the multimodal realization of the warning sentences.

These results suggest that, as with Gesture E, the production of Gesture B may play a structural role in the high-level warning pattern. Proportionally, we observed that 80% of participants who use non-manual gestures also employ Gesture B.

6.2.3 Gesture H (Head Forward)

A separate mixed-effects logistic regression model was fitted for Gesture H (Head Forward) to examine the influence of the stimulus on its production. The model, which utilized a binomial distribution, revealed extremely high inter-individual variability, with a large random effect of participant (Variance = 14.56, SD = 3.816).

Regarding the fixed effects, none of the stimuli reached statistical significance in predicting the production of Gesture H. These findings confirm the results of the previous experiment, in that its consistent presence across conditions points to a fundamental structural role in marking the pragmatic function of a warning, regardless of the specific danger level or the explicit presence or absence of a warning call. This is further supported by the observed proportions: among speakers who produced non-manual gestures (10 out of 13 participants), 70% used Gesture H, indicating its high prevalence across all warning conditions.

6.2.4 Statistical Analysis of Gesture Co-occurrence (Pattern E-H-B)

A log-linear analysis was conducted on a three-way contingency table for gestures E, H, and B in Experiment B. The goal was to test for significant interactions among these gestures. The analysis revealed a statistically significant three-way interaction ($p = 0.03$). This result indicates that the co-occurrence of the three gestures is not a matter of chance; instead, the relationship between any two gestures is significantly dependent on the presence or absence of the third.

The significance of the three-way interaction in Experiment B, in contrast to the non-significant finding in Experiment A, is a crucial outcome. This suggests that the composition of the stimulus set—specifically, the inclusion of a larger number of high-warning contexts—may be a key modulator of the multimodal pattern. The more pronounced and frequent elicitation of high-level warning contents seems to have strengthened the gestural pattern, pushing the gestures to co-occur as a unified, interdependent triad. Namely, taken together, the data indicate that Gesture B tends to be inhibited in low-level warning contexts, whereas high-level warning scenarios are more likely to elicit its production. This provides compelling evidence that the E-H-B pattern is not merely an incidental phenomenon, but a robust and context-sensitive multimodal structure that becomes more integrated and cohesive in response to higher levels of communicative urgency and perceived danger.

6.2.5 Statistical comparison between experiments A and B

To investigate whether overall gesture frequency differed between experiments, we fitted a generalized linear mixed-effects model (GLMM) with a negative binomial distribution, using

Experiment as a fixed effect and Participant ID as a random intercept to account for inter-individual variability. The model revealed a significant difference between Experiments A and B (Estimate = -0.56 , SE = 0.27 , $z = -2.05$, $p = 0.04$), indicating that, overall, participants in Experiment B produced fewer gestures of all types compared to those in Experiment A. The inclusion of Stimulus as an additional fixed effect in a more complex model slightly reduced the significance of the group difference (Estimate = -0.50 , SE = 0.28 , $z = -1.79$, $p = 0.074$), while no individual stimulus reached statistical significance. These results suggest that the observed difference in gesture frequency is primarily driven by the experimental context, specifically the intended pragmatic nuance elicited, rather than by the specific stimuli presented. Notably, random effect estimates confirmed substantial inter-individual variability, reflecting consistent differences in the overall propensity to gesture among participants.

Regarding the E–H–B gesture pattern, the analysis confirmed that high-warning-related utterances elicit higher gesture production compared to contexts with lower pragmatic urgency. The mixed-effects logistic regressions for key gestures E, B, and H revealed uniformly low baseline probabilities in Experiment A (ranging from $\approx 0.7\%$ to 3%). No statistically significant differences emerged between Experiments A and B for any gesture type. Inter-participant variability was consistently substantial, while stimulus-related variability was either negligible (E, H) or modest (B). In conclusion, according to our analysis, E–B–H is a recurrent 'warning pattern' common to both experiments.

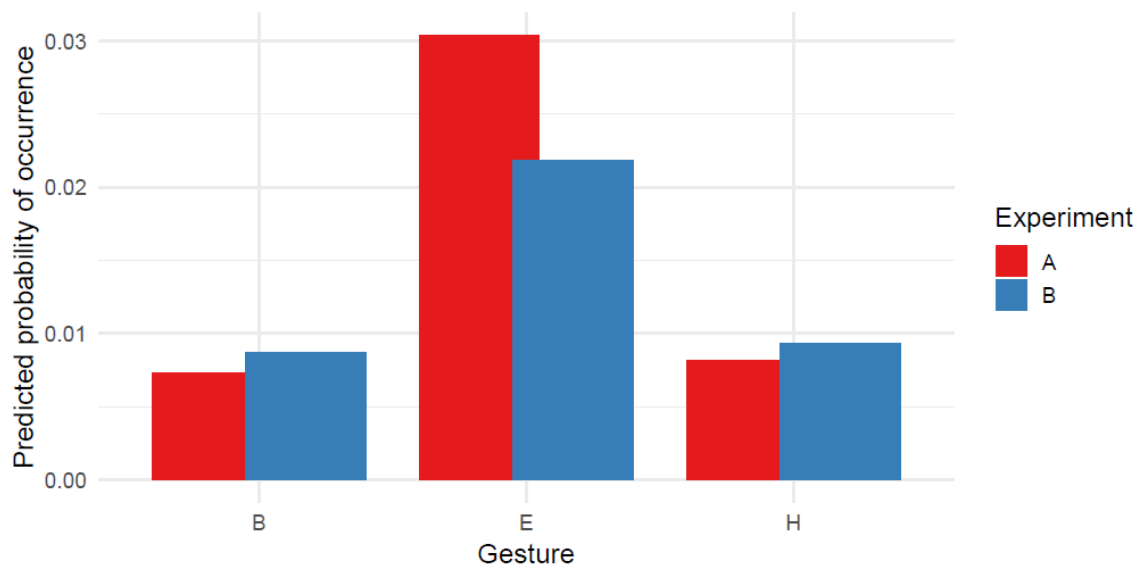


Figure 5: Consistent E–H–B gesture pattern across Experiments A and B

Figure 5 shows the predicted probabilities of producing gestures E, H, and B in Experiments A and B, based on mixed-effects logistic regression models. The plot illustrates that probabilities are comparable across the two experiments, with no gesture showing a significant difference, indicating that the E–H–B pattern is consistently present in both groups. Thus, despite individual variability, the probability of producing the E–H–B pattern remained stable across both experiments. This indicates that it is the context, rather than individual differences, that determines the presence of these gestures. Inter-individual variability accounts instead for differences in frequency, that is, in the number of gestures produced, but not in the types of gestures selected. In other words, what varies across participants is the tendency to produce or inhibit the gesture, rather than the emergence of the pattern itself. Consistent with this interpretation, when the level of danger increases, speakers either increase the production of warning gestures or show a lower degree of inhibition.

Figures 6 and 7 provide a detailed comparison between sentences with an explicit warning call and presentative warnings without an explicit warning call. The stimuli with an explicit warning call (Figure 6) corresponded to situations associated with a higher level of danger (e.g., the risk of hitting a child crossing the street, or running a red light), whereas the stimuli in Figure 7 included both high-risk contexts and low-risk contexts:

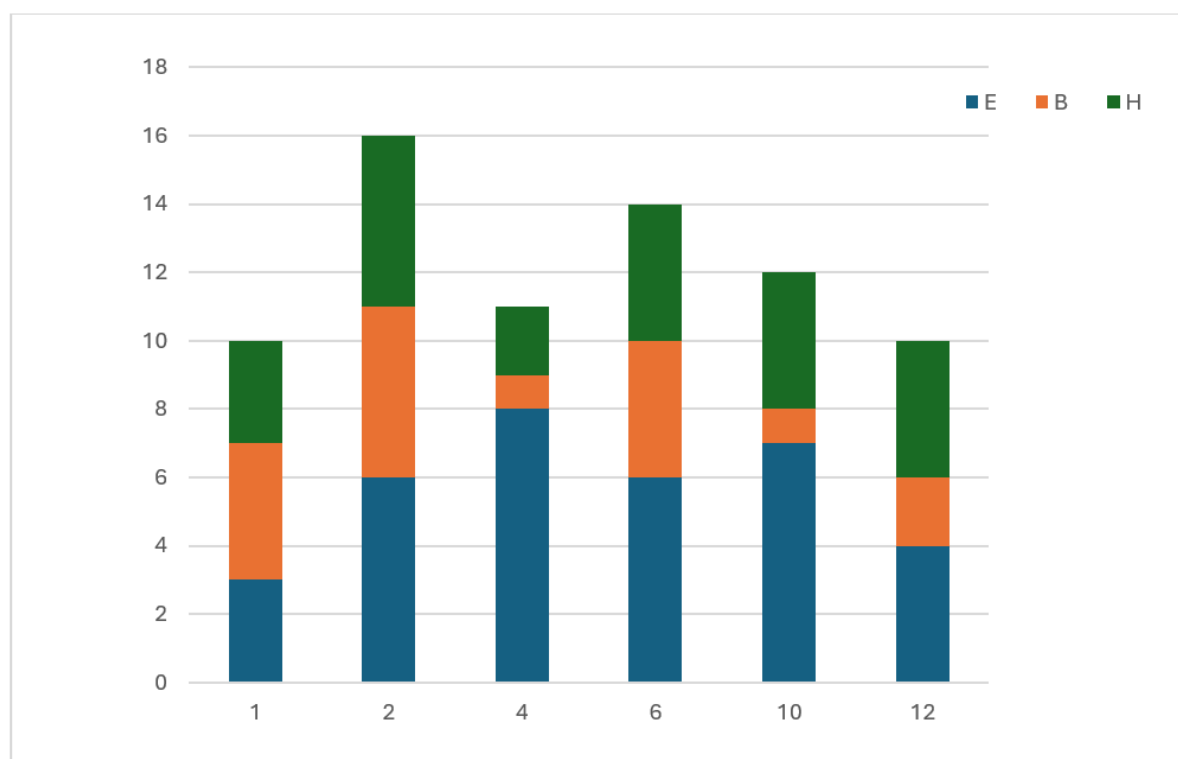


Figure 6: E, H and B Gesture Production in Explicit Warning Stimuli

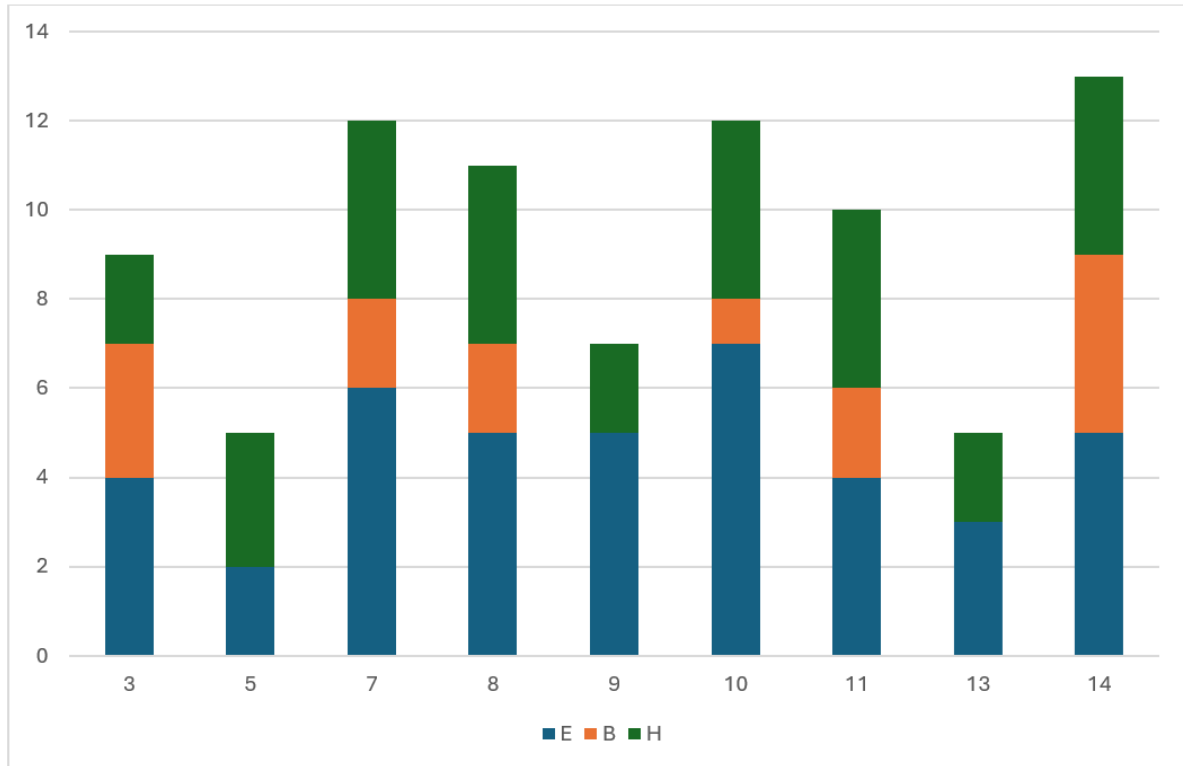


Figure 7: E, H and B Gesture Production in Silent Imperative Warning Stimuli

Figures 6 and 7 plot the stimuli along the horizontal axis (x-axis), while the vertical axis (y-axis) represents the number of gestures produced. As shown by Figures 6 and 7, the distribution of gestures is very similar across stimuli, regardless of the presence of an explicit warning call. This suggests that an overt warning call does not significantly affect the overall pattern of gesture production, reinforcing the idea that a silent imperative is consistently realized in the syntax of this type of sentence.²³

The only difference relates to the production pattern of gesture B. The graphs show that, while gesture B consistently present in the contexts shown in Figure 6, it is missing in some contexts in Figure 7 (Stimuli 5, 9, and 13). As we have seen, gesture B is highly sensitive to the warning level conveyed by the stimulus. In particular, it seems that B is easily suppressed in low-warning contexts.

²³ Note also that no discrepancies are observed in Stimuli 1 and 12, which follow the same pattern despite the presence of an attention call preceding the presentative warning call: *Hey! Watch out! It is red!*

6.3 Experiment C (Controls)

6.3.1 Questions

In the neutral *wh*-question condition, we first examined whether the total number of gestures varied across stimuli using a negative binomial mixed-effects model. The model revealed no significant effect of stimulus on gesture count, with all coefficients non-significant ($p > 0.2$). The dispersion parameter was extremely high (7.76×10^9), indicating either severe overdispersion or highly unbalanced data, likely due to a predominance of zero-gesture observations. Some coefficients were effectively zero (e.g., for Stimuli 4, 5, and 12), suggesting that in those contexts no gestures beyond zero occurred, preventing the estimation of stimulus effects.

At the level of individual gestures, generalized linear mixed-effects models were fitted for Gestures E, F (Furrowed eyebrows), Sh (Head shake), and Side (Head to the side). For Gesture E, no effect of stimulus was observed ($z = -0.148$, $p = 0.882$), and similarly for Gesture F ($z = 0.282$, $p = 0.778$), Sh ($z = 1.579$, $p = 0.114$), and Side ($z = 0.000$, $p = 1.000$). Gestures B and H could not be modeled because these gestures were never produced across any neutral *wh*-question stimulus. Overall, these results indicate that neutral *wh*-questions do not elicit a systematic gestural pattern, and that the E–H–B pattern observed in warning sentences is not present in these canonical, neutral contexts.

6.3.2 Assertions

The negative binomial mixed-effects model exhibited convergence warnings and a non-positive-definite Hessian, indicating potential estimation issues. The dispersion parameter was extremely large (1.26×10^7), suggesting severe overdispersion or highly unbalanced data, likely due to a predominance of zero-gesture observations. Several stimulus coefficients were effectively zero (e.g., Stimulus 3), implying that in those contexts no gestures occurred beyond zero, preventing reliable estimation of stimulus effects. Overall, these results indicate that the total number of gestures does not systematically vary across the neutral declarative stimuli.

In this dataset, consisting entirely of canonical/neutral declarative sentences, no regular multimodal pattern was observed. The Poisson mixed-effects model for the total number of gestures revealed an extremely large dispersion parameter (7.76×10^9), indicative of severe overdispersion or highly unbalanced data with a preponderance of zeros. Several stimulus coefficients were effectively zero (e.g., Stimulus4, Stimulus5, Stimulus12), suggesting that in those contexts no observation contained more than zero gestures, making reliable estimation

impossible. Turning to individual gesture models, estimation was not possible for B and H because these responses were never produced. Gesture E was rare (Intercept log-odds = -1.03 , $p = 0.083$), with no significant stimulus effects; similar patterns of rarity or model instability were found for other gestures such as F (Furrowed eyebrows), Sh (Head Shake), Side (Head to the Side), and N (Nod). Overall, no evidence emerged for the E–H–B configuration or any other recurrent gesture pattern, confirming that these gestures do not arise in neutral contexts and supporting the view that the E–H–B sequence is tied to specific pragmatic conditions rather than general gestural behavior.

6.4 Interim conclusions

Based on the analysis above, we can conclude that E–B–H is indeed a pattern, characterizing warnings as opposed to affirmative and interrogative sentences. Furthermore, its presence is not significantly affected by the presence or absence of a warning call, as no statistically significant difference was found between experiments A and B. Lastly, there may be differences among speakers in the number of gestures they produce, but not in the type. Another important observation is that although the number of gestures correlates with the severity of the danger presented in the context, these differences are not significant. All these observations support our hypothesis, as we will discuss in section 7.

7 Conclusive remarks

Let's now briefly consider an issue that is not easily measurable but is important for understanding the phenomena in question: alignment. The alignment of the gestural pattern, E, B, and H, with phonological production and syntactic representation, is one of the most significant cues for understanding the underlying structure. Alignment is a key argument for viewing the prosodic and phonological realization of a sentence as a multimodal output of the sensorimotor interface and therefore as part of grammar. This point was also explored and studied in our previous research on special questions – Giorgi & Dal Farra (2019), Petrocchi (2022), and Giorgi & Petrocchi (2024).²⁴

²⁴ A reviewer points out that our results are only correlations and do not prove causality. This is certainly true; however, repeating experiment A – namely, experiment B – under slightly different conditions, along with the results of the control experiment, offers additional evidence supporting our hypothesis. It should always be remembered that, by definition, a hypothesis is still a hypothesis and must be subject to falsification by further experimental evidence in favor of a better one.

As already noted in Sections 6.1 and 6.2, whenever more gestures of the pattern occur in the same sentence, they are realized simultaneously. A closer look reveals that, in presentative warning sentences, in 100% of the cases, the activation of E, H, and B either precedes or is at least simultaneous with utterance onset. When an explicit warning call is present, by contrast, activation of the gestures is consistently time-locked to the call.²⁵

Finally, in cases where an interjection precedes the warning call, e.g., *Ehi. Attento. C'è un bambino* (Hey. Watch out. There is a child), activation still occurs on the warning call and never on the interjection *Ehi*. This was observed in 95% of the cases overall (26 stimuli, 2 X 13 participants).

These observations, although only qualitative, are very significant because they confirm that the gestural pattern we noted is specifically associated with warning expressions. The interjection (INT), as discussed in Giorgi (2023), constitutes an autonomous part of the discourse and is linked to the rest of the utterance through a silent discourse head, resulting in a structure such as the following one:²⁶

(21) [[_{Int-P} INT] DIS₀ [[_{Appr-P} APPR₀ [*imperative*]] DIS₀ [(*che*) *description of danger*]]]

The interjection is used to capture the listener's attention, making the warning more effective, but it is not part of the warning itself. Coherently with the syntactic structure, the gestural pattern is activated only afterward, in sync with the warning, i.e., with the apprehensive projection.

We can now draw some conclusions that address the research questions in section 4. Regarding the first question – is there a typical gestural pattern for warnings? – we found that a specific gestural pattern characterizes warning expressions. This pattern includes the following gestures: raised eyebrows, head leaning forward, and body leaning forward, clearly setting warnings apart from affirmative and interrogative sentences. The occurrence of this pattern is sensitive to the severity of the danger: more severe dangers trigger more gestures.

²⁵ In order to check on this issue, we checked the ELAN annotations of the participant's productions. For reasons of space, here we can only briefly describe the results.

²⁶ A reviewer inquires about the different feature specifications of the various discourse heads in this representation. It is unclear whether discourse heads must necessarily be characterized by features of some kind; so far, we do not have an answer to this question. This remains an interesting empirical issue for further research.

In particular, raised eyebrows and body leaning forward are produced by nearly all subjects in high-danger scenarios. Across the two experiments, considering all 36 participants, raised eyebrows (E) were the most frequently produced gesture, with an estimated total occurrence of 94%. Head forward (H) was produced by approximately 78% of participants, while body forward (B) was observed in about 68% of participants. Despite some inter-individual variability and differences between the two experimental contexts, the E–H–B pattern remains a robust and recurrent feature across participants.

Regarding the second issue – the role of the explicit warning call in the stimuli – the results from experiment B, where 6 stimuli included an overt warning call, show that there is no significant difference among the stimuli. This indicates that the type and number of gestures in a warning are independent of whether an explicit warning call is present. Additionally, considering the cases in experiment A where subjects spontaneously added a warning call, we can conclude that the hypothesis presented in section 4 about the existence of an implicit warning call, even when it is not lexicalized, becomes more plausible.

In Experiment A, 14 out of 23 participants spontaneously produced additional material beyond the presentative warning sentence that the experimenter asked them to repeat. Participants uttered *attento* (Watch out) alongside the presentative warning sentence in 42.9% of cases, with this occurring across stimuli without any apparent preference for high-warning contexts.²⁷ Finally, in 14.3% of cases, participants produced an interjection functioning as an attention call, such as *ehi* (hey), before the presentative warning sentence and/or preceding the warning call that comes before the presentative warning sentence.

Our conclusion is further supported by our observations on alignment: The trigger for the gestural pattern always occurs to the left of the presentative sentence, regardless of whether a warning call is lexicalized. Moreover, as mentioned earlier, the gestural pattern does not just activate as soon as possible, in that, if a preceding attention call – *hey!* – is present, the pattern only activates *after* the attention call. In other words, its activation is a direct effect of the implicit or explicit warning call, i.e., of the silent imperative.

Finally, regarding question 3, we have demonstrated that there is a significant difference in the gesture inventory between warnings on one side and both assertions and questions on the other, indicating that the pattern identified is typical of this type of speech act.

There is one additional question that, however, can only be partially addressed here, as it is

²⁷ In some instances, alternatively, participants spontaneously added imperative forms such as *guarda!* (look!).

mostly outside the scope of this work. Why are these specific gestures triggered in warnings? Given that their presence and association are statistically significant, why these particular ones and not others? Here, we can only offer some speculation on this matter.

We observed that within the E–H–B warning pattern, head leaning forward, H, is not linked to the danger's severity, as it occurs across all danger stimuli, indicating it is a basic gesture in these contexts. Raised eyebrows, E, were the most common, with 94% of speakers showing them, especially in high-warning contexts. Body-forward movement, B, is the most easily inhibited, as it tends to vanish in low-warning contexts. We can hypothesize that forward postural gestures of the head and torso represent the anticipation of an upcoming action, specifically one aimed at actively responding to an imminent threat.²⁸

Looking at our data, raised eyebrows and forward head movement appear more stable, functioning as consistent cues within the warning pattern, whereas body posture seems to retain a more pronounced postural quality, at least in spoken languages. This aligns with previous observations in the literature, according to which changes in body position can be interpreted by the interlocutor as an indicative attitude of the speaker, that is, for example, the decision to act upon a danger, for instance, by physically moving in response to a threat. (Noorozi et al., 2021; De Gelder, 2006). This stronger indexicality, which could underlie the very origin of the gesture, may explain why body-forward movement is used selectively, emerging primarily in high-danger contexts where urgency and imminent action need to be conveyed.

We also observed manual gestures in addition to postural gestures. In particular, we observed pointing gestures that were consistently aligned with the explanation of the danger; that is, the pointing gesture was used to indicate the threat that the speaker intended to alert the interlocutor to, aligning with the word specifying the danger, such as ‘step’ in the sentence *There is a step!* However, since manual gestures are not the focus of the present paper, we did not analyze them here.

In conclusion, we propose enriching the syntactic representation so that certain language

²⁸ In general, little is known about head gestures beyond nodding, tilting, and shaking, and torso gestures are the least studied category, as noted by Noorozi et al. (2021). In Italian Sign Language (LIS), it is already known that the signer's shoulders mark the imaginary boundary from which a future action is projected in front of the signer's shoulders or a past action behind them (Bertone 2011; Branchini, Mantovan 2020:142). Finally, it has been argued that raised eyebrows are associated with the need to open the eyes to better attend to an unexpected event and are typically linked to surprise (Ekman 2003). See also Noordewier and van Dijk (2018) for an analysis of the dynamic unfolding of surprise in facial expressions.

features, traditionally seen as extra-syntactic and managed in a separate grammar module, i.e., pragmatics, are now incorporated into syntax. This approach is supported by the consistency of the observed patterns, which are far from being idiosyncratic, and by the strong alignment with the independently motivated syntactic structure. An integrated model like this offers clear and testable predictions, as we discussed throughout the paper, opening the way for further research in language and human cognition.

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